

RONIT ROY CHOUDHARY

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Profile

Bioinformatician and Computational Biologist with expertise in cancer transcriptomics, NGS analysis, and deep learning-based predictive modelling. Experienced in biomarker discovery and integrative analysis using GEO and TCGA datasets, with a current research focus on AI-driven approaches for genomic and transcriptomic data analysis

Experience

Research Trainee

The Himalayan Centre for High-throughput Computational Biology (HiChiCoB), CSIR-IHBT
January 2026 – Present | Palampur, India

Supervisor: [Dr. Ravi Shankar](#)

- Developed an automated NGS-based variant calling and antibiotic resistance profiling pipeline for *Mycobacterium leprae*, improving reproducibility and reducing manual analysis time.
- Designed and implemented an end-to-end Hi-C data analysis workflow using the Juicer framework for chromatin interaction mapping.
- Built a machine learning model to predict Hi-C contact maps from RNA-seq expression profiles, exploring transcriptome–chromatin architecture relationships.
- Performed quality control, alignment, variant calling, and downstream genomic analysis using industry-standard bioinformatics tools.

Research Intern

Department of Surgical Oncology, Institute of Medical Sciences (IMS), Banaras Hindu University (BHU)
May 2025 – July 2025 | Varanasi, India

Supervisor: [Dr. Manoj Pandey](#)

- Conducted integrative transcriptomic analysis of head and neck cancer using GEO and TCGA datasets.
- Identified differentially expressed genes (DEGs) and constructed PPI and gene–miRNA networks using STRING, Cytoscape, and NetworkAnalyst.
- Performed functional enrichment analysis (GO, KEGG) and pathway mapping to identify key oncogenic signaling mechanisms.
- Conducted Kaplan–Meier survival analysis and Cox proportional hazards modelling to evaluate prognostic biomarkers.
- Integrated multi-layer bioinformatics results to propose candidate biomarkers for precision oncology applications.

Education

M.Sc. Bioinformatics and Computational Biology

Devi Ahilya Vishwavidyalaya | CGPA: 8.62 | Expected July 2026

B.Sc. Biotechnology

Bharati Vidyapeeth | CGPA: 8.25 | Nov 2021 – June 2024

Skills

- **Programming:** Python (Pandas, NumPy, Biopython), SQL, C++, Bash
- **Bioinformatics:** NGS analysis, Variant calling, Differential expression analysis, Hi-C analysis (Juicer), Genome annotation
- **Machine Learning & Deep Learning:** TensorFlow, PyTorch, CNNs, Transformers, Mamba, Predictive modelling
- **Bioinformatics Tools:** Cytoscape, STRING, NetworkAnalyst, GSEA, DAVID, BLAST
- **Databases:** GEO, TCGA, NCBI, UniProt, DrugBank, PDB
- **Statistical Analysis:** Kaplan–Meier analysis, Cox regression, Log2FC, FDR correction

Certifications

- Data Analyst Skill Path – Udemy
- Computational Protein Modelling and Analysis – NPTEL (SWAYAM)
- Introduction to Microsoft Excel – Coursera
- Excel Skills Job Simulation – JPMorgan Chase & Co. (Forage)
- iON Career Edge – Young Professional – TCS
- Next-Generation Sequencing for Influenza & SARS-CoV-2: From Sample Collection to Analysis – WHO

Conferences & Presentations

- **Selected for Presentation – [COPM 2026](#) (International Conference)**
Silesian University of Technology, Poland | April 29, 2026
 - Presented *integrative transcriptomic analysis of TSCC identifying cross-population molecular signatures and regulatory networks*
 - Aligned with theme: From Genome to Treatment: AI & Precision Medicine
 - Abstract published in official conference proceedings

Publications

- “*Integrative Transcriptomic Analysis Identifies Shared Coding and Non-Coding Biomarkers in TSCC Across Diverse Populations*” (Manuscript under review)
Authors: **Ronit Roy**; Monika Rajput; Manoj Pandey
- “*Preoperative NLR and PLR as Robust Prognostic Biomarkers for Oral Squamous Cell Carcinoma*” (Manuscript under review)
Authors: Tarun Kumar; Mandar Tilak; Akshay Kantha; Shayak Roy; Monika Rajput; **Ronit Roy**; Manoj Pandey